

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

22nd November 2024

SP4 - Pensions and Other Benefits

**Time allowed: 3 Hours 15 Minutes (09.30 – 12.45
Hours)**

Total Marks: 100

Q. 1)

- I. List the difference between a defined benefit (DB) pension scheme and a defined contribution (DC) pension scheme. [4]

A large multinational corporation, ABC Corp, has both Defined Benefit (DB) and Defined Contribution (DC) pension schemes for its employees. Due to changing market conditions and regulatory requirements, ABC Corp is considering restructuring its pension arrangements to better align with its long-term financial and strategic goals.

As a consultant specializing in pension schemes, you have been tasked with advising ABC Corp on the restructuring of its DB and DC pension schemes.

- II. Develop a comprehensive plan outlining the steps ABC Corp should take to restructure its pension arrangements, considering the following factors:
- The current financial status of the DB scheme, including any surplus or deficit.
 - The impact of restructuring on current and future employees' retirement benefits.
 - Regulatory and legal considerations, including compliance with pension laws and regulations.
 - Communication strategy for informing employees about the changes and addressing any concerns.
 - Options for restructuring, such as merging the DB and DC schemes, converting the DB scheme to a DC scheme, or offering enhanced benefits in the DC scheme.
 - Implementation, monitoring, review and adjustments

[15]

[19 Marks]

- Q. 2)** You are an Actuary of a defined benefit scheme of a company. You have set up the assumptions to value the liabilities for the current year funding valuation. A note on assumptions has been shared with the Trustees. The Trustees of this scheme have deliberated and asked you to provide your inputs on their following comments:

- I. "Given the current economic climate with low interest rates and market volatility, we are concerned about the appropriateness of our current discount rate assumption for valuing our pension liabilities. Could you provide your recommendations on how we should adjust our discount rate?" [5]
- II. "Given the recent rise in inflation rates, we are worried that our current salary increase and benefit escalation assumptions might not be sufficient. How should we adjust these assumptions to account for the expected higher inflation?" [5]
- III. "Our workforce demographics have been changing significantly, with higher turnover rates and an aging population. What adjustments should we make to our turnover and retirement age assumptions to better reflect these trends?" [5]
- IV. "Our goal is to minimize volatility in contribution levels to ensure stable funding for the pension scheme. What recommendations do you have for setting actuarial assumptions that support this objective?" [5]

Provide suitable replies to the Trustee to address their concern.

[20 Marks]**Q. 3)**

- I. Discuss various considerations while distributing the surplus of the defined benefit scheme. [4]

A company sponsors a defined benefit pension scheme that has accumulated a surplus. The company is considering using the surplus to provide enhanced benefits to certain groups of members, such as early retirees or long-serving employees.

- II. Discuss the potential actuarial, legal and administrative considerations that the company should take into account when making this decision. [10]

The trustees of the schemes are not in favour of using the surplus for selected members. The alternative has been suggested to use the surplus to implement a pension increase exchange exercise, where members can exchange future pension increases for a one-time cash payment.

- III. Discuss the potential benefits and risks for the company, the pension scheme, and its members, addressing financial, legal, and reputational impacts, administrative and funding effects, member outcomes, legal and regulatory requirements, communication strategies, long-term sustainability, and governance considerations. [20]

[34 Marks]

Q. 4)

- I. How should the trustees monitor the strength of the sponsor? [5]

The XYZ Steel Pension Scheme was sponsored by XYZ, UK, a subsidiary of ABC Steel, which is facing significant financial difficulties due to market conditions and the global steel industry crisis.

- II. Given that the sponsor of the defined benefit pension scheme is experiencing financial difficulties, how should actuaries adjust their valuation and funding advice using techniques like Value at Risk (VaR) and other quantitative methods to ensure the scheme's sustainability, and outline specific actions the trustees may take based on this advice. [12]

Given the weak sponsors' covenant and the Actuary advice, the trustees of the pension scheme evaluated various options and ultimately chose to transfer the liabilities to an insurer to guarantee the benefits for all scheme members. This decision was based on the insurer's strong financial health and the ability to secure long-term stability for the members' benefits.

- III. When trustees decide to transfer the liabilities of a discontinued pension scheme to an insurer to guarantee benefits, describe various factors that must be considered to ensure the security and adequacy of the benefits. [10]

[27 Marks]
